

# Probability Theory and Statistics

## Lecture 16

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## Theorem (Law of Total Expectation )

*Let  $X$  and  $Y$  be random variables such that  $\mathbb{E}(|X|)$  and  $\mathbb{E}(|Y|)$  are finite. Then*

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X)).$$

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## Theorem (Law of Total Expectation with a Partition)

*Let  $A_1, \dots, A_n$  be a partition of  $\Omega$  such that  $\mathbb{P}(A_i) > 0$  for all  $i$ . If  $\mathbb{E}(|Y|) < \infty$ , then*

$$\mathbb{E}(Y) = \sum_{i=1}^n \mathbb{E}(Y|A_i)\mathbb{P}(A_i).$$

## Theorem (Law of Total Expectation, Discrete Case)

Let  $X$  be a discrete random variable with possible values  $\{x_1, \dots, x_n, \dots\}$  such that  $\mathbb{P}(X = x_i) > 0$ . If  $\mathbb{E}(|Y|) < \infty$ , then

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## Theorem (Law of Total Expectation, Continuous Case)

Let  $X$  be a continuous random variable with  $\mathbb{E}(|Y|) < \infty$ , and let  $f_X$  denote its density function. Then

$$\mathbb{E}(Y) = \int_{-\infty}^{\infty} \mathbb{E}(Y|X = x)f_X(x)dx.$$

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**Exercise 2:** Toss a fair coin. If it is head, then we take a number from  $Y \sim \text{Exp}(1/2)$ , otherwise we take a number from  $Z \sim N(2, 1)$ . What is the expectation of the final number?

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**Exercise 3:** We can choose two different boxes,  $A$  and  $B$ . The probability of choosing  $A$  is  $\frac{1}{3}$  and the probability of choosing  $B$  is  $\frac{2}{3}$ . In the  $A$  box, there are 3 red and 5 blue balls. In the  $B$  box, there are 4 red and 3 blue balls. From the chosen box we take out 3 balls. (Without replacement.) What is the expectation of the number of blue balls that are discarded?



# Reminder

## Theorem (Law of Total Probability for the Discrete Case)

*Let  $X$  be a discrete random variable and  $A$  an event. Then*

$$\mathbb{P}(A) = \sum_{k \in \text{Ran} X} \mathbb{P}(A|X = x) \mathbb{P}(X = k).$$

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## Theorem (Law of Total Probability for the Continuous Case, or “Continuous LTP”)

*Let  $X$  be a continuous random variable and  $A$  an event. Then*

$$\mathbb{P}(A) = \int_{-\infty}^{\infty} \mathbb{P}(A|X = x) f_X(x) dx,$$

*where  $f_X$  is the density of  $X$ .*

## Exercises 2

**Exercise 1:** We choose an  $x$  number uniformly from 2 to 5. After the choose a  $y$  number from  $x$  and  $x^2$ . What is the probability that  $y < 20$ ?

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# Convolution of random variables

Convolution:  $X + Y$

We have seen:

## Theorem

*If  $X \sim \text{Poi}(\lambda)$  and  $Y \sim \text{Poi}(\mu)$  are independent random variables, then  $X + Y \sim \text{Poi}(\lambda + \mu)$ .*

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## Theorem

*If  $X \sim \text{Bin}(n, p)$  and  $Y \sim \text{Bin}(m, p)$  are independent random variables, then  $X + Y \sim \text{Bin}(n + m, p)$ .*

# Convolution of Normal Distributions

We haven't seen the case of two independent normal distributions.

## Theorem

*If  $X \sim N(\mu_1, \sigma_1^2)$  and  $Y \sim N(\mu_2, \sigma_2^2)$  are independent random variables, then  $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ . Moreover  $cX \sim N(c\mu_1, c^2\sigma_1^2)$*

(Proof)

# Exercise for the convolution of normal distributions

**Exercise:** Let  $X \sim N(2, 1^2)$  and  $Y \sim N(3, 2^2)$  be independent random variables. If  $V := X + Y$  and  $W := X - Y$ , then

- what is  $\text{cov}(V, W)$ ?
- what is the linear regression of  $V$  on  $W$ ?



